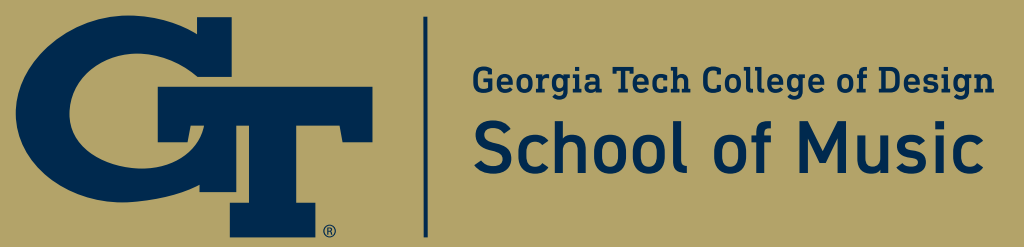


Beyond Static Benchmarks: A Post-Leaderboard Evaluation Paradigm for Generative Music and Creative AI

Alexander Lerch
Music Informatics Group, Georgia Institute of Technology



Evaluation Crisis in Generative Music

The current state of evaluation in generative music is **methodologically fragmented**, relies on **ambiguous validation objectives**, and builds on **systematically misaligned metrics** [1]. This has led to a crisis in the field, where the following failures are observed:

- **Validity failure:**
 - Utilized metrics are *unreliable proxies* and fail to measure musical qualities.
- **Leaderboard failure:**
 - Single-score optimization leads to *overfitting* and is *insufficient for assessing a multi-dimensional cultural artifact*.
- **Comparability failure:**
 - *Different metrics, feature representations, and reference data* prevent meaningful comparisons across models.
- **Reproducibility failure:**
 - *Closed-source representations and evaluation scripts* and proprietary data hinder reproducibility.
- **Governance failure:**
 - *Ad-hoc, fragmented evaluation protocols* indicate lack of standardization.

Goals of the Proposed Framework

- **Community-driven ecosystem governance:**
 - coordinated by the community for the community.
 - metrics and evaluation protocols have to *evolve with the field*.
- **Multi-dimensional evaluation:**
 - acknowledging the *multi-dimensional nature of music*.
- **Interpretability:**
 - utilizing *interpretable metrics* and features where possible.
- **Domain adaptability:**
 - acknowledging *variety of use cases* through curated reference benchmark profiles (RBPs).
- **Modularity & extensibility:**
 - modular software architecture allows for easy *integration of new features and evaluation methods*.
- **Reproducibility & transparency:**
 - versioned *open source* software with comprehensive documentation and *auditable logging*.

Governance Structure

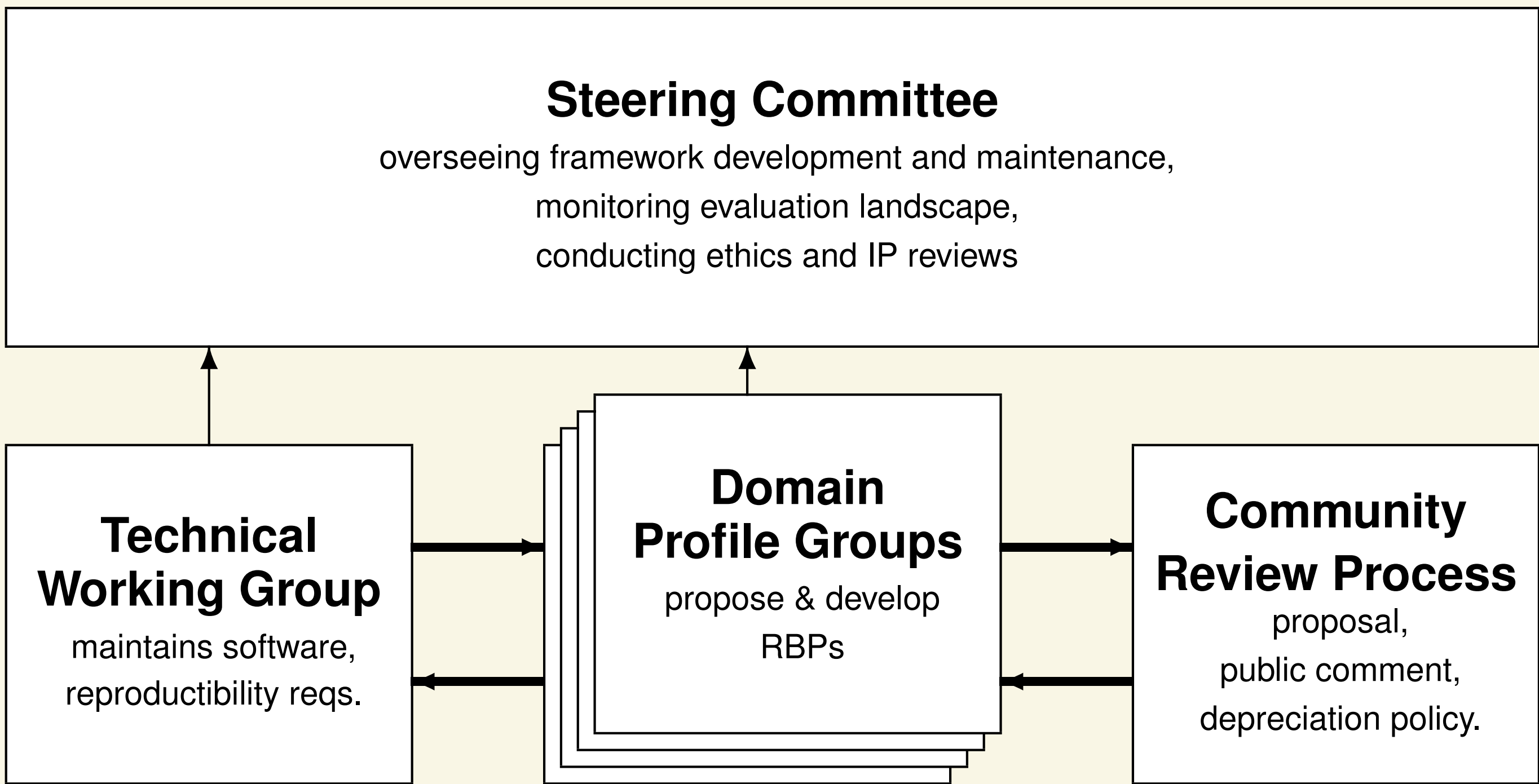


Figure: Proposed governance structure

Cross-Domain Applicability

- modular architecture $\Rightarrow$ easily transferable to other domains of generative AI by **replacing feature representations**; candidates are
 - *Visual Art*
 - *Text Generation*
 - *Game Content Generation*

Software Architecture

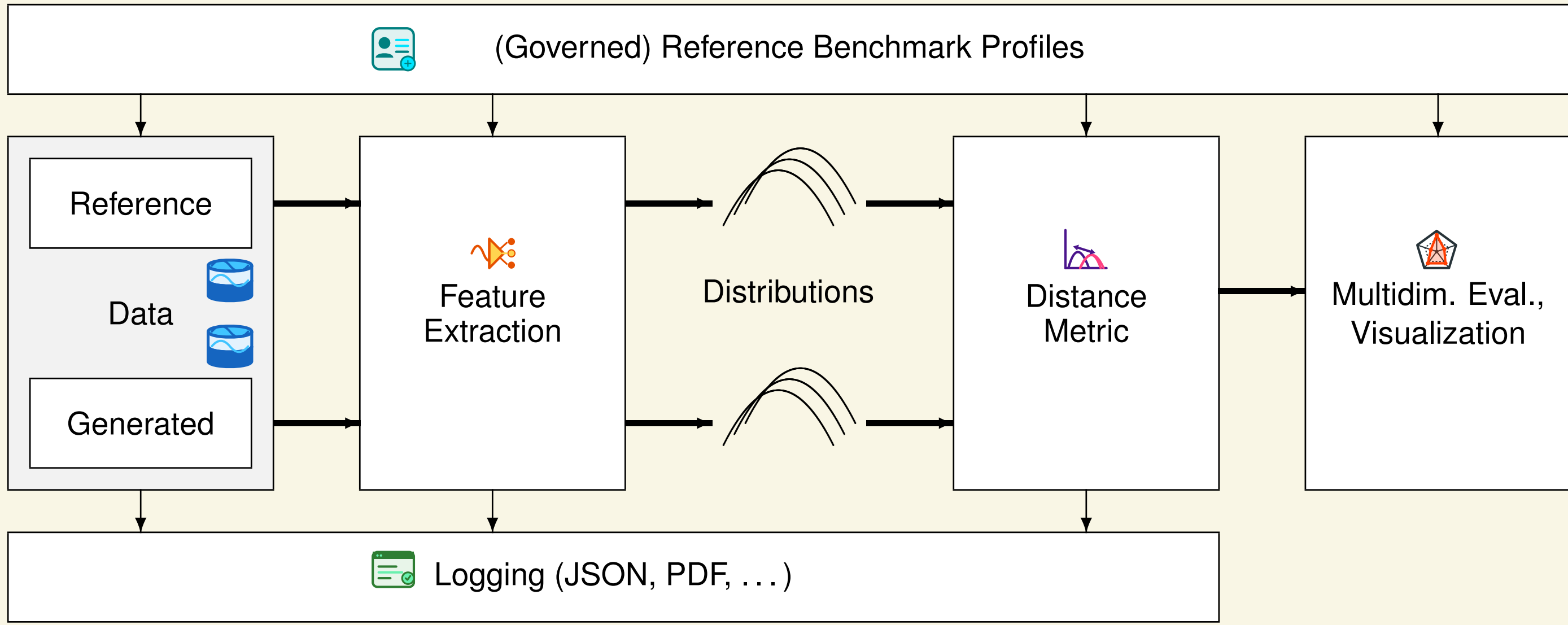


Figure: Proposed framework architecture for the evaluation of generative music.

- **Reference Benchmark Profiles (RBPs):**
 - use-case specific combination of
 - *reference data*
 - *feature representations*
 - *distance metrics*
 - *reporting format*
- **Standardized reference data:**
 - publicly available, documented datasets as controllable open references.
- **Standardized feature representations:**
 - musical qualities
 - audio fidelity
 - controllability
 - originality
- **Standardized distance metrics:**
 - distributional distance metrics (Wasserstein, KL, MMD, etc.)
 - traditional evaluation metrics (accuracy, precision, recall, F1, etc.) for classification tasks
 - task-specific metrics as needed
- **Multi-dimensional results & visualization:**

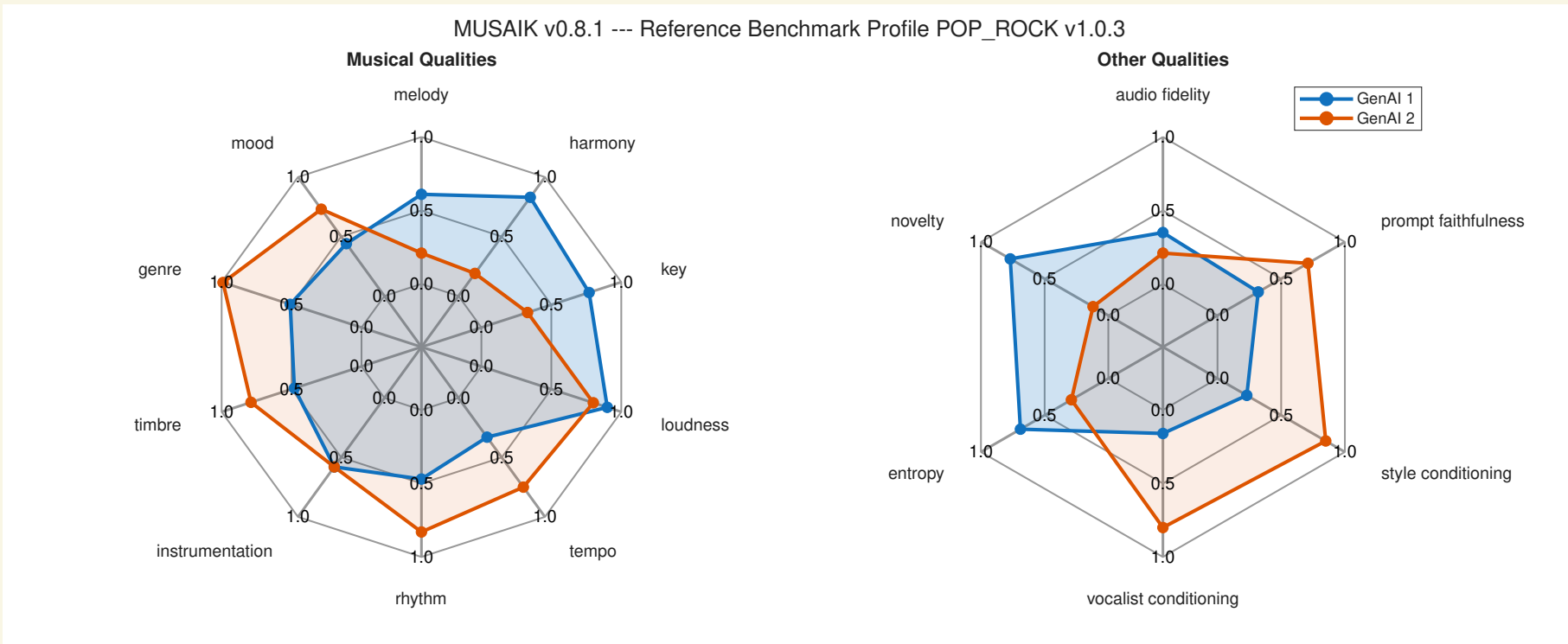


Figure: Mockup of aggregated result presentation.

- **Logging:**
 - auditable logs (machine & human readable formats) with RBP, version info, test data, etc.

Limitations

- **Missing subjective evaluation protocols**
- **Black-box evaluation:**
 - no evaluation of the underlying model architecture/capabilities
 - no distinction between human contributions and fully machine-generated output
- **Focus on output qualities:**
 - no usability or user experience evaluation
 - no evaluation other characteristics, such as explainability, bias, resource usage, ethical use of data, etc.

References

- Alexander Lerch, Claire Arthur, Nick Bryan-Kinns, Corey Ford, Qianyi Sun, and Ashvala Vinay. Survey on the Evaluation of Generative Models in Music. *ACM Computing Surveys*, 58(4):99:1–99:36, October 2025.